CSMFAB000000000113-Vol-12

Scale 3 — Small Amateur (5–50 kg): Complete Analysis

Series: CSMFAB000000000113 | **Volume:** Vol 12 of 29 | **June 2026 Notes:** Research class; Karman line borderline; second stage needed

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1. Configuration Summary

Parameter	Value
Spear mass range	5–50 kg
Representative mass (m_rep)	27.5 kg
Launch depth	25 m
Vessel radius (R = 0.2d)	5 m
Rope length (L_rope)	25.5 m
Rope angle (alpha)	11.3 deg
Geometric efficiency	98.1%
Number of vessels (n)	6–8
Representative n	7

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2. Velocity and Energy Budget

Stage 1 — Underwater Archimedean screw exit velocity:

```
v_screw = 22 m/s  (saturated exit for spear at depth 25 m)
```

Stage 2 — Smart Rope delta-v:

```
delta_v = n * J_rope * eta_v / m_rep
        = 7 * 1001 * 0.981 / 27.5
        = 250 m/s
```

Total exit velocity:

```
v_exit = v_screw + delta_v = 22 + 250 = 272 m/s
```

= **Mach 0.79** at sea level (343 m/s sound speed)

Kinetic energy at exit:

```
KE = 0.5 * 27.5 * 272^2 = 1017.0 kJ
```

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3. Trajectory Analysis

3.1 Vacuum Ballistic (theoretical max, no atmosphere)

```
H_vacuum = v_exit^2 * sin^2(42 deg) / (2g)
         = 272^2 * 0.447 / 19.62
         = 1.7 km
```

3.2 Atmospheric Trajectory (realistic, with drag)

```
H_atmosphere = H_vacuum * 0.65 (empirical correction for atmospheric drag losses)
             = 1.1 km
```

3.3 Maximum Range (flat Earth, optimal angle ~42 deg)

```
R_range = v_exit^2 * sin(2*42 deg) / g = 7.5 km
```

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4. Orbital Velocity Gap Analysis

LEO orbital velocity: 7,800 m/s Exit velocity achieved: 272 m/s **Delta-V gap: 7528 m/s = 96.5% of orbital velocity remaining**

Second stage required for orbital attempt

For orbital attempt from 1.1 km apogee: Required second stage delta-v:

```
dv_2 = sqrt(v_orbital^2 + 2*mu*(1/r_apogee - 1/R_earth)) - v_apogee
     = 7746 m/s  (approximate; see Vol 17 for full analysis)
```

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5. Vessel Specifications

Minimum vessel length: 8 m (must carry 25 m rope + crew + power) **Minimum vessel tonnage:** 1.4 tonnes (reaction during rope release) **Vessel displacement during launch:** 0.550 cm (acceptable) **Required sea state:** Beaufort 3 or below

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6. Rope Specifications

Parameter	Value
Rope type	Smart Rope (CSMFAB000000000022 marine grade)
Segment size	25 mm cube
Segments per rope	1019
Mass per rope	8.76 kg
Total rope system mass (n=7)	61.34 kg
Pre-tension force	13750 N
Release force (max, per rope)	50,040 N

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7. Safety Margins

Component	Design Load	Material Limit	Safety Factor
Spear body (ZrB2-SiC)	7479 N · s / 0.0937 s	850 MPa	>50
UHMWPE rope fibers	50,040 N (max)	4250 MPa * 7.85e-7 m² = 3,336 N/strand	2.0 (30 strands)
Rope attachment to spear	350280 N total	ZrO2 joint: 1300 MPa	>100

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8. Practical Build Notes

Research class; Karman line borderline; second stage needed

Materials sourcing: - ZrB2-SiC spear: Treibacher Industrie AG (Austria), $280-320/kg powder; flash-sinter in-house - UHMWPE fiber: DSM Dyneema SK99, $35-55/kg; available in 100-500m spools - KNbO3-BaTiO3:

TDK 2025 lead-free catalog, $180/kg in actuator form - Vessels: charter or own BFRP-hull vessels (available for offshore research)

Citations (Vol-12)

- CSMFAB000000000022 V2.0, Smart Rope specs, CSM 2026
- CSMFAB000000000109 V2.0, Archimedes screw exit velocity data, CSM 2026
- Vol 02 (this series), Smart Rope energy analysis
- Vol 05 (this series), Multi-vessel geometry
- Vol 06 (this series), Depth optimization data
- Bate, R.R., Mueller, D.D., White, J.E., Fundamentals of Astrodynamics, Dover 2020
- Anderson, J.D., Introduction to Flight, McGraw-Hill 2017
- SpinLaunch technical publications, 2024-2025

End Vol-12 | CSMFAB000000000113 | CSM Safe Pod Engineering Company | June 2026